

Editorial: Special Issue on Statistical Methods and Techniques for Analyzing Spatial and Temporal-Spatial Data

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The Università degli Studi di Siena was the site of the sixth international meeting on quantitative methods for applied sciences during 19–20 September 2002. Hosted by the Dipartimento di Metodi Quantitativi, this meeting focused on statistical methods and techniques for analyzing spatial and temporal-spatial data. I was privileged to participate in this meeting at the invitation of its organizers, and I came away from it quite stimulated by the range of topics and diversity of statistical applications that were discussed during these two days. In this issue of *Environmental and Ecological Statistics*, I am pleased to include five articles that grew out of oral presentations at the meeting and were subsequently prepared by the authors and examined by external referees.

Two of these papers present novel applications of the bootstrap resampling procedure. In “Multivariate bootstrap confidence regions for abundance vector using data depth”, Di Battista and Gattone develop a bootstrap method to obtain an approximating empirical distribution of a multivariate abundance vector. An application is made to estimate the abundance vector of a benthic community in Lake Lesina, Italy. Manly applies the bootstrap to ameliorate small-sample bias following two-phase adaptive sampling in the article entitled “Using the bootstrap with two-phase adaptive stratified samples from multiple populations at multiple locations.”

Two-stage sampling provides the context for Fattorini and Pisani’s contribution “Variance decomposition in two-stage plot sampling: theoretical and empirical results.” The surprising results of this decomposition when estimating abundance stand in clear contrast with results in the environmental literature that have been reported earlier. I speculate that this paper will be of great interest to resource managers who use a first stage of sampling to select spatial units over the landscape where the resource is located.

The challenges posed by sampling environmental resources that are spatially dispersed are also addressed in “Optimal spatial sampling schemes for environmental surveys” by DiZio, Fontanella, and Ippoliti. Adopting a fractal characterization of spatial variation, these authors propose a fractal-based criterion and fractal spatial simulated annealing to determine optimal sampling locations.

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Barabesi and Marcheselli provide interesting and useful insights into the design-based performance of the common ratio and regression estimator in “Design-based ranked set sampling using auxiliary variables.” The sampling strategies that they examine offer potentially large gains in estimation efficiency when sampling populations that display a nonlinear relationship between the variate of interest and the auxiliary variate.

I wish to thank the organizers for sponsoring this meeting in Siena in 2002, and for assisting me in getting papers for submission to *Environmental and Ecological Statistics*. As always, I am indebted to the many referees who reviewed the submitted manuscripts and offered helpful suggestions to the authors.